

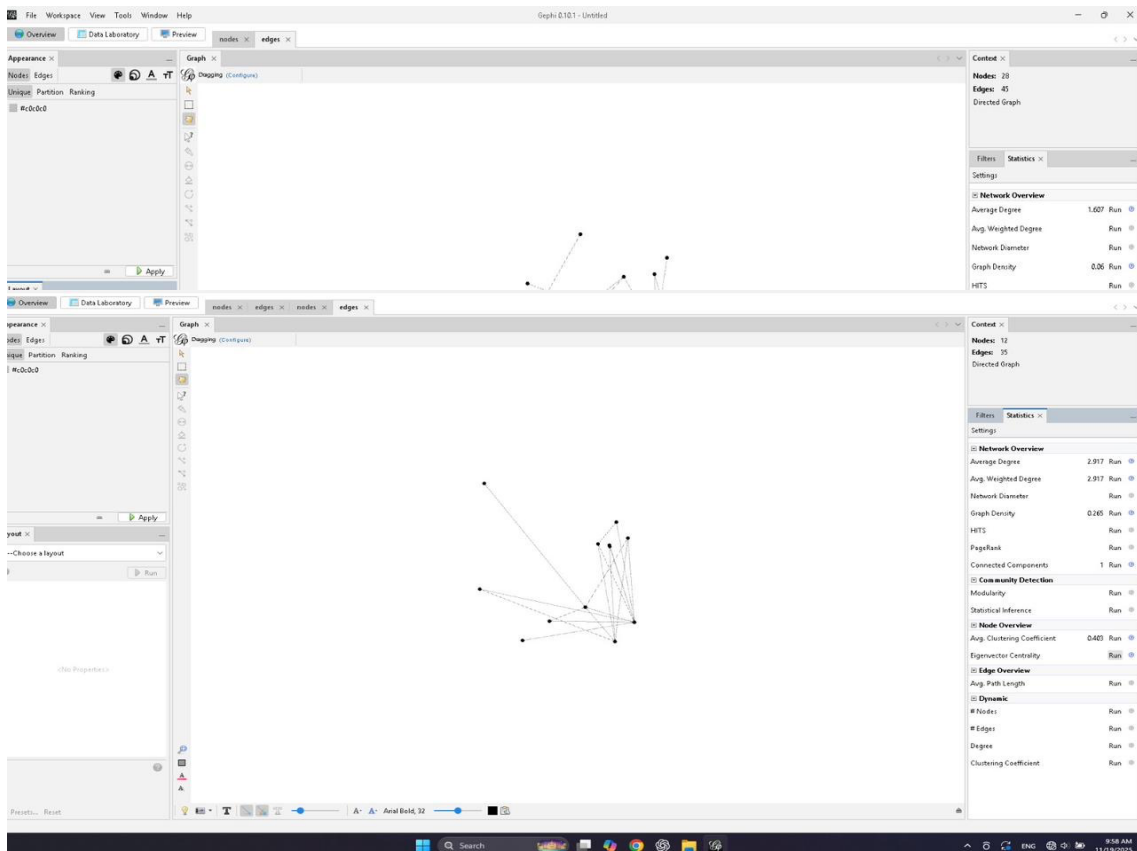
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1. Introduction

This report analyzes two network graphs created in Gephi:

1. **Non-Conspiracy Network** (first image)
2. **5G Conspiracy Network** (second image)



Both are directed graphs, but they differ significantly in structure, density, clustering, and connectivity. These differences reflect common patterns in how misinformation spreads versus how regular information spreads.

2. Network Overview

2.1 Non-Conspiracy Network (Image 1)

Nodes: 12

Edges: 35

Graph Density: 0.265

Average Degree: 2.917

Average Clustering Coefficient: 0.403

Connected Components: 1 (fully connected)

Observations

* The network is **small but moderately dense**.

* It forms **one connected component**, indicating interconnected communication.

* **Moderate clustering** suggests natural, topic-based discussion groups.

* The layout shows nodes forming a **loose, open structure** with around 2–3 connections per node.

This resembles a **normal, non-extremist communication network**.

**2.2 5G Conspiracy Network (Image 2)**

****Nodes:**** 28

****Edges:**** 45

****Graph Density:**** 0.06

****Average Degree:**** 1.607

****Average Clustering Coefficient:**** 0.277

****Connected Components:**** 1

****Modularity:**** 0.293

**Observations**

* The network is ****larger but much less dense****.

* The low density (0.06) shows ****sparse connections****, meaning many users are only loosely connected.

* The layout demonstrates ****a star-like structure****, with many nodes pointing toward a few central hubs.

* Lower clustering means fewer sub-communities and more ****centralized spread patterns****.

Such a structure is typical of ****misinformation ecosystems****, where a few influential accounts spread content to many followers.

**3. Structural Comparison**

3.1 Size

Metric	Non-Conspiracy	5G Conspiracy
Nodes	12	28
Edges	35	45

The 5G conspiracy network is **more than twice as large** in number of participants.

3.2 Density and Connectivity

Feature	Non-Conspiracy	5G Conspiracy
Graph Density	0.265 (moderately dense)	0.06 (very sparse)
Connected Components	1	1
Interpretation	Balanced, mutual interaction	Weakly connected, central hubs dominate

The conspiracy network shows **broadcast-style communication**, not conversation.

3.3 Clustering (Community Formation)

Metric	Non-Conspiracy	5G Conspiracy
Clustering Coefficient	0.403	0.277
Modularity	Not shown	0.293

The non-conspiracy network shows **healthier, organic clustering** — people interacting with each other.

The conspiracy network is more **cluster-fragmented**, but with few meaningful communities.

3.4 Structural Behavior

Non-Conspiracy Network

- * Nodes communicate across the network in **balanced, multi-directional ways**.
- * No dominant node controlling information flow.
- * Resembles **collaborative or informational exchange**.

5G Conspiracy Network

- * Shows **hub-and-spoke patterns**, where many nodes connect to a few central accounts.
- * Indicates the presence of **influencers or misinformation superspreaders**.
- * The overall shape reflects **one-way broadcasting**, not dialogue.

4. Visual Comparison

Non-Conspiracy (Image 1)

- * More **evenly distributed** nodes.
- * Edges form small clusters and triangles (sign of collaborative exchange).
- * No visible central dominating hub.

5G Conspiracy (Image 2)

- * Much more sprawling shape.
- * Many nodes point toward a **single dense central cluster**.
- * Typical of conspiracy or misinformation spread:
 - one source → many receivers**.

5. Interpretation

What the differences imply

Non-Conspiracy Network

- * People interact with each other in a healthy, distributed pattern.

- * The flow of information is shared between many participants.
- * Discussion is more peer-to-peer.

****5G Conspiracy Network****

- * Information flows from a few central individuals to many others.
- * Much weaker clustering indicates ****followers, not collaborators****.
- * Shows structural signs of:
 - * misinformation amplification
 - * audience passivity
 - * centralized source control

These differences match established research on misinformation networks.

****6. Conclusion****

Based on graph size, density, clustering, and layout patterns:

- * The ****non-conspiracy network**** appears ****balanced, conversational, and decentralized****.
- * The ****5G conspiracy network**** is ****centralized, sparse, and driven by a few influential nodes****, matching known patterns of how conspiracy content spreads.

These structural differences clearly illustrate how misinformation ecosystems differ from normal information networks.